Non-smooth Optimization
Subgradient Method

1 Regularization

In order to perform "shrinkage" of solution in some form (like zeroing out some weights), a common technique is to add regularization term to the objective that pushes solution towards some simplified form. This section covers such methods.

The most common example is LASSO regularization:

$$\frac{1}{N}\sum_{i=1}^{N} l(y_i, \langle w, x_i \rangle) + \lambda \|w\|_1 \to \min_w$$

Or, more generally:

$$f(w) + \lambda \|w\|_1 \to \min_w \tag{1}$$

To see why this leads to sparsification, we can consider the problem with L1 constraint, and then show that it is equivalent to the original one. The constrained problem is:

$$\begin{aligned} \min_w \quad & f(w) \\ \text{subject to} \quad & \|w\|_1 \le \eta. \end{aligned} \tag{2}$$

We can write KKT conditions for the optimum of such task. There exists $(\hat{w}, \hat{\lambda})$, such that:

$$\begin{aligned} & \hat{w} \in \operatorname{Arg}\min_w \ \mathcal{L}(w, \hat{\lambda}), \\ & \|\hat{w}\|_1 \le \eta \qquad \hat{\lambda} \ge 0 \\ & \hat{\lambda}\big(\|\hat{w}\|_1 - \eta\big) = 0. \end{aligned} \tag{3}$$

Suppose that the original task (1) has solution w^*. Then, if we substitute (w^*, λ) and $\eta = \|w^*\|_1$, into KKT conditions (3), we can see that they are all satisfied. Therefore, w^* is also a solution to the task (2). So, task (1) and (2) are equivalent.

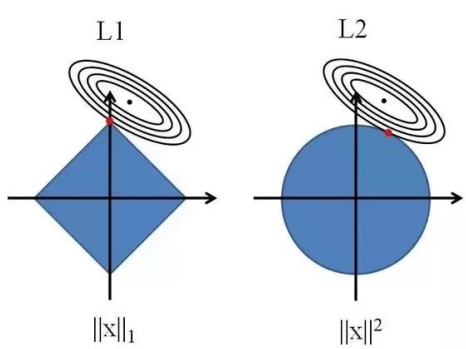

Figure 1: LASSO sparsification visualization.

A common way to intuitively explain the sparsity induced by constrained task (2) is the diagram Fig. 1. Due to L1 constraint having the form of rhombus, the levels of the function $f(w)$ are "likely" to intersect it on one of its corners. Therefore, some coordinates get zeroed out. L2 constraint, on the other hand, can be intersected anywhere and does not induce sparsity.

One extension of this approach is so-called **group sparsity**. It can be used in cases where we want to zero out entire groups of weights, not weights individually. This may be beneficial in terms of saving computation. For example, zeroing out some rows of the matrix allows throwing away some input dimensions and considering a smaller matrix.

Suppose we split our weights into G groups:

$$w = [w_1, w_2, \ldots, w_G]$$

Group sparsity regularization term has the form:

$$\|w\|_{2,1} := \sum_{g=1}^{G} \|w_g\|_2$$

Intuitively, we consider the group as a whole with L2 norm, and penalize L1 norm of the groups. In fact, when each group contains only a single element, $\|w\|_{2,1} = \|w\|_1$.

Fig. 2 shows an example when 3 coordinates (w_1, w_2, w_3) are split into groups: (w_1, w_2), (w_3) and constraint has the form:

$$\sqrt{w_1^2 + w_2^2} + |w_3| \leq 1$$

The figure has a form of a spinning top. We can see that its "corners" are either $z = 0, \quad x^2 + y^2 = 1$ or $x = y = 0, \quad |z| = 1$, so the solution is "likely" to intersect it on those points.

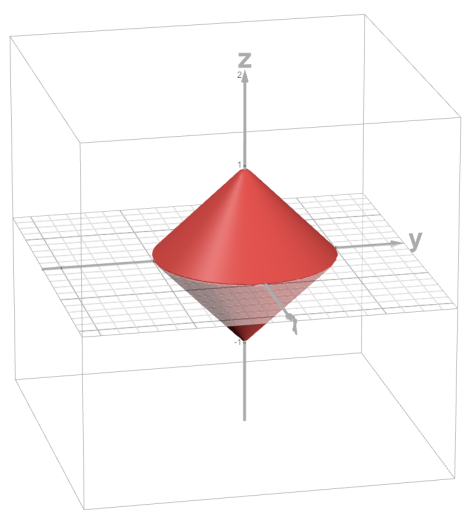

Figure 2: Constraint $\sqrt{x^2 + y^2} + |z| \leq 1$

If we have a weight matrix W and we want to obtain a low-rank solution, which is equivalent to zeroing out some singular values, we can use a constraint which is L1 norm on the singular values. It is also called the **nuclear norm**:

$$\lambda \|W\|_* := \lambda \sum_i \sigma_i(W)$$

Another example is the constraint to enforce a binary structure on the weights:

$$\lambda \sum_i \min(|w_i - 1|, |w_i + 1|)$$

The regularization function is visualized in Fig. 3. Notably, it is *not convex*.

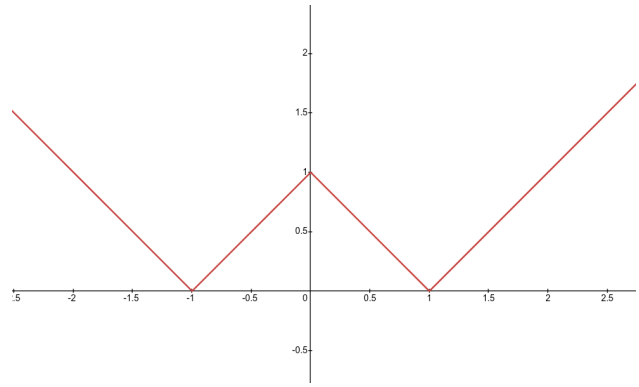

Figure 3: Function $\min(|x-1|, |x+1|)$

2 Subgradient

2.1 Definition

A common feature of all aforementioned regularization terms is that they are, in general, *not smooth*. For example, $\|w\|_1$ is not smooth at any points where at least one of the coordinates is zero.

We would like to optimize the convex functions with such non-smooth regularization. Therefore, we consider a general case when convex and non-smooth function f is given:

$$f(x) \to \min_x, \quad f \in C, \quad f \notin C^1, \quad f \text{ is convex}$$

Subdifferential of function f at point x is defined as:

$$\partial f(x) := \{z \mid f(y) \geq f(x) + \langle z, y - x \rangle \quad \forall y\}$$

Intuitively, subdifferential is a set of all tangent planes that lower bound the function f. Any point $z \in \partial f(x)$ is called a **subgradient**. Fig. 4 shows an example of a function and its subgradients for different points. In particular at non-smooth point (0, 0) there are many possible subgradients.

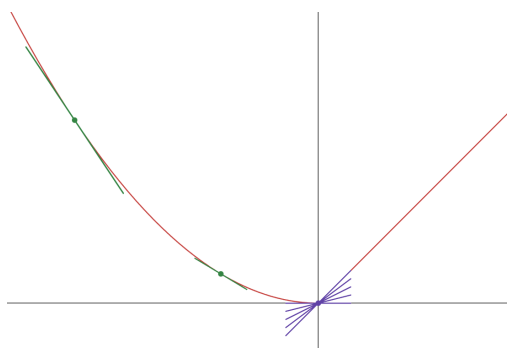

Figure 4: Subgradient illustration.

Statement 1. *Subdifferential is a convex set.*

Proof. Consider 2 subgradients $z_1, z_2 \in \partial f(x)$. By definition of subgradient, we have:

$$f(y) \geq f(x) + \langle z_1, y - x \rangle$$
$$f(y) \geq f(x) + \langle z_2, y - x \rangle$$

Multiplying first equation by α and second by $(1 - \alpha)$ and then adding 2 equations together gives us:

$$f(y) \geq f(x) + \langle \alpha z_1 + (1 - \alpha) z_2, y - x \rangle$$

Since this holds true for all y, all z_1, z_2 and for all $\alpha \in [0, 1]$, subdifferential is a convex set. □

Consider a function $f(x) = |x|, \quad x \in \mathbb{R}$ as an example. Its subdifferential is:

$$\partial f(x) = \begin{cases} \{1\}, & x > 0, \\ \{-1\}, & x < 0, \\ [-1, 1], & x = 0. \end{cases}$$

By definition of subdifferential, $z \in \partial f(0) \iff |y| \geq |0| + z(y - 0)$ for all y. Hence, z values from -1 to 1 suffice. Also, we can see that $[-1, 1]$ is a convex set.

Statement 2. *Consider a convex non-smooth continuous function f. Then:*

$$x \in \operatorname*{Arg\,min}_z \; f(z) \iff 0 \in \partial f(x)$$

Proof.

$$0 \in \partial f \iff f(y) \geq f(x) + \langle 0, y - x \rangle \iff f(y) \geq f(x) \;\; \forall y \iff x \text{ — minimum}$$

□

2.2 The Subgradient Method

The **subgradient method** is a minimization algorithm with the following update rule:

$$x_{k+1} = x_k - \alpha_k z_k$$
$$z_k \in \partial f(x_k)$$

At every step, we take *any* subgradient of the function and perform update based on it. We do not control which subgradient is being chosen.

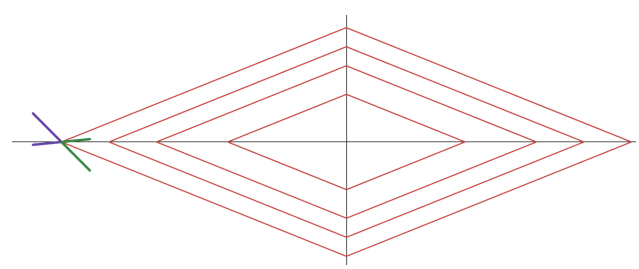

Figure 5: Levels of function $f(x_1, x_2) = \alpha|x_1| + |x_2|$. Purple lines show subgradient directions at the corner of level set and green directions are negative subgradients. At points of non-convexity, some negative subgradients may not lead to descent of the function.

The subgradient method does not have "descent" in its name because the update does not necessarily decrease the function. An example is function $f(x_1, x_2) = \alpha|x_1| + |x_2|$, for which at some point $(\hat{x}_1, 0)$ negative subgradient direction may lead to increase of the function. It is visualized in Fig. 5.

Let us prove some results on convergence of the subgradient method. We will consider a convex function with Lipschitz bound $f \in C_L^{0,0}$:

$$|f(y) - f(x)| \leq L\|y - x\| \ \ \forall y, x$$

Iterations can be written as:

$$x_{k+1} = x_k - \alpha_k z_k, \quad z_k \in \partial f(x_k)$$

Let us denote the optimal point as x_* and the solution as f_*. Now, we derive inequalities:

$$\|x_{k+1} - x_*\|^2 = \|x_k - x_* - \alpha_k z_k\|^2 = \|x_k - x_*\|^2 - 2\alpha_k \langle z_k, x_k - x_* \rangle + \alpha_k^2 \|z_k\|^2 = \ldots =$$
$$= \|x_0 - x_*\|^2 - 2\sum_{i=0}^{k} \alpha_i \langle z_i, x_i - x_* \rangle + \sum_{i=0}^{k} \alpha_i^2 \|z_i\|^2$$

Now, at each point x_i, we know that $z_i \in \partial f(x_i)$. Therefore, we know that:

$$f(x_*) \geq f(x_i) + \langle z_i, x_* - x_i \rangle \implies \langle z_i, x_i, -x_* \rangle \geq f(x_i) - f_*$$

Substituting that into the previous expression, we get:

$$\|x_{k+1} - x_*\|^2 \leq \|x_0 - x_*\|^2 - 2\sum_{i=0}^{k} \alpha_i \left(f(x_i) - f_*\right) + \sum_{i=0}^{k} \alpha_i^2 \|z_i\|^2$$

Let us denote with $\overline{x}_k$ the point from previous iterations $i = 1, 2, \ldots, k$ on which the function value $f(x_i)$ is minimal: $f(\overline{x}_k) \leq f(x_i) \ \ i = 1, 2, \ldots, k$. Then, we can write:

$$2\left(\sum_{i=0}^{k} \alpha_i\right)\left(f(\overline{x}_k) - f_*\right) \leq 2\sum_{i=0}^{k} \alpha_i \left(f(x_i) - f_*\right) \leq \|x_0 - x_*\|^2 + \sum_{i=0}^{k} \alpha_i^2 \|z_i\|^2 - \|x_{k+1} - x_*\|^2 \leq$$
$$\leq \|x_0 - x_*\|^2 + \sum_{i=0}^{k} \alpha_i^2 \|z_i\|^2$$

Suppose that the initial point x_0 is within the R radius of the optimal point x_*: $\|x_0 - x_*\| \leq R^2$. Also, we can show that due to Lipschitz constant we can show that subgradients always satisfy: $\|z_i\| \leq L$. Combining this allows us write the final convergence bound:

$$\boxed{f(\overline{x}_k) - f_* \leq \frac{\|x_0 - x_*\|^2 + \sum_{i=0}^{k} \alpha_i^2 \|z_i\|^2}{2\left(\sum_{i=0}^{k} \alpha_i\right)} \leq \frac{R^2 + L^2 \sum_{i=0}^{k} \alpha_i^2}{2\left(\sum_{i=0}^{k} \alpha_i\right)}} \tag{4}$$

Let us now consider different choices of learning rates α_k.

1) Constant learning rate $\alpha_k = h$ gives:

$$\frac{R^2 + h^2(k+1)L^2}{2(k+1)h} = \frac{R^2}{2(k+1)h} + \frac{hL^2}{2} \xrightarrow{k \to \infty} \frac{L^2 h}{2}$$

So, choosing smaller constant gives better final result, however it slows down the convergence in early stage.

2) Normalized update

$$\alpha_k = \frac{h}{\|z_k\|} \geq \frac{h}{L}$$

Then, we get:

$$\frac{R^2 + (k+1)h^2}{2(k+1)\frac{h}{L}} = \frac{LR^2}{2h(k+1)} + \frac{Lh}{2} \xrightarrow{k \to \infty} \frac{Lh}{2}$$

Notably, this is better than the constant learning rate due to L having power 1 instead of 2.

3) Learning rate schedule that satisfies:

$$\sum_{i=1}^{\infty} \alpha_i^2 \leq +\infty \quad \sum_{i=1}^{\infty} \alpha_i = +\infty$$

A common choice of such schedule is:

$$\alpha_i = \frac{h}{(i+1)^\tau}, \quad \tau \in \left(\frac{1}{2}, 1\right]$$

For $\tau = 1$, the partial sum $\sum_{i=0}^{k} \alpha_i$ behaves like an integral:

$$\int_1^k \frac{h}{x} dx = h \log k$$

Plugging that into the convergence bound Eq. 4, we get a bounded numerator and unbounded denominator growing like $\log k$. So, the convergence speed is:

$$f(\overline{x}_k) - f_* = O\left(\frac{1}{\log k}\right)$$

So, unlike previous 2 schedules, this choice of learning rates gives convergence. However, it is very slow due to slow growth of logarithm.

4) Learning rate choice such that:

$$\sum_{i=1}^{\infty} \alpha_i^2 = +\infty \quad \sum_{i=1}^{\infty} \alpha_i = +\infty \quad \frac{\sum_{i=1}^{k} \alpha_i^2}{\sum_{i=1}^{k} \alpha_i} \xrightarrow{k\to\infty} 0$$

A common choice is again schedules of form:

$$\alpha_i = \frac{h}{(i+1)^\tau}, \quad \tau \in \left(0, \frac{1}{2}\right]$$

In particular, for $\tau = 1/2$ the numerator grows as $\log k$ while denominator as $\sqrt{k}$. So, we get a convergence speed.

$$f(\overline{x}_k) - f_* = O\left(\frac{\log k}{\sqrt{k}}\right) = \tilde{O}\left(1/\sqrt{k}\right) \tag{5}$$

Due to very slow growth of logarithm, it is sometimes omitted and the speed is written with $\tilde{O}$. It can be shown that convergence speed in Eq. 5 is optimal for the subgradient method. This is slower than for standard gradient descent (Tab. 1), which is expected as we have less restrictions on the optimized function (it no longer has to be smooth).

Method	Function class	Convergence rate
GD	$f \in C_L^{1,0}$, convex	$O(1/k)$
Subgradient	$f \in C_L^{0,0}$, convex	$\tilde{O}(1/\sqrt{k})$

Table 1: Comparison of GD and the subgradient method

3 Subdifferential Calculus

3.1 Properties

This section covers basic properties of subdifferentials. We consider a function $f : E \to \mathbb{R}$. We call function f **subdifferentiable** at point x if $\partial f(x) \neq \emptyset$ and subdifferentiable on set E if it is subdifferentiable at any point $x \in E$.

Statement 3.

$$f \textit{ subdifferentiable on convex } E \implies f \textit{ is convex on } E$$

Proof. Consider 2 points $z_1, z_2 \in E$. Due to convexity of E: $\alpha z_1 + (1-\alpha) z_2 \in E$. Since f is subdifferentiable at z, we write definition for z_1 and z_2.

$$f(z_1) \geq f(\alpha z_1 + (1-\alpha) z_2) + \langle g_z, (1-\alpha)(z_1 - z_2) \rangle$$
$$f(z_2) \geq f(\alpha z_1 + (1-\alpha) z_2) + \langle g_z, \alpha(z_2 - z_1) \rangle$$

Multiply the first equation by α, second one by $(1-\alpha)$ and add them. scalar products cancel out and we get the definition of convexity for function f:

$$\alpha f(z_1) + (1-\alpha) f(z_2) \geq f(\alpha z_1 + (1-\alpha) z_2)$$

□

Now we're interested in the inverse relation. Generally, it is not true. However, the problems arise only on boundaries. The next property shows that.

Statement 4.

$$f \textit{ convex on } E,\ x \in \textit{int} E \implies \partial f(x) \neq \emptyset$$

Next, we relate ordinary differentiability and subdifferentiability.

Statement 5.

$$f \textit{ convex and differentiable at } x \in \textit{int} E \implies \partial f(x) = \{\nabla f(x)\}$$

Proof. From differentiability of the convex function we know the property:

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \implies \nabla f(x) \in \partial f(x)$$

Now, let us prove that there are no other points. Suppose there is a subgradient $s \in \partial f(x),\ f \neq \nabla f(x)$. Then, we apply the definition of subgradient for point $x + tv$.

$$f(x + tv) \geq f(x) + \langle s, tv \rangle$$

Dividing by t we get:

$$\frac{f(x + tv) - f(x)}{t} \geq \langle s, v \rangle$$

Now for the limit $t \to 0+$ we get a directional derivative for direction v:

$$\langle \nabla f(x), v \rangle \geq \langle s, v \rangle \implies \langle \nabla f(x) - s, v \rangle \geq 0$$

Since x is in the interior, all directions v are available. Therefore, $\nabla f(x) = s$ and we get a contradiction. □

If subdifferential consists of only one subgradient, then this subgradient is exactly the gradient, and function is differentiable at that point.

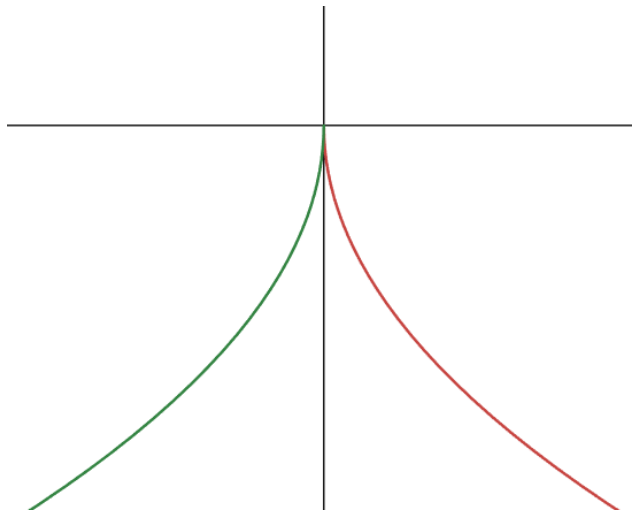

Figure 6: Example of addition property

Statement 6.

$$f \textit{ convex},\ \partial f(x) = \{z\} \implies f \textit{ differentiable at } x$$

3.2 Computing Subdifferentials

Now, we cover typical ways of computing subdifferentials.

$$\begin{gathered} f(x),\ c > 0 \\ \partial(cf)(x) = c \times \partial f(x) \end{gathered} \tag{6}$$

$$\begin{gathered} f : E \to \mathbb{R},\ g : G \to \mathbb{R},\ x \in E\ \cap\ G \\ \partial(f+g)(x) \supseteq \partial f(x) + \partial g(x) \\ f, g \text{ convex},\ E\ \cap\ \text{int}G \neq \emptyset \implies \\ \implies \partial(f+g)(x) = \partial f(x) + \partial g(x) \end{gathered} \tag{7}$$

Here we consider set scaling as element-wise scaling and addition of 2 sets as all possible additions between element from first set and element from second set.

Notably, in property (7) the "problem" arises only at boundaries between 2 sets. For example, consider functions (Fig. 6):

$$f(x) = -\sqrt{x} \quad x \geq 0 \quad \partial f(0) = \emptyset$$
$$g(x) = -\sqrt{-x} \quad x \leq 0 \quad \partial g(0) = \emptyset$$

However, if we consider a function $(f+g)$ at their intersection, which is only $\{0\}$, since we have only 1 point, any vector is subgradient and therefore:

$$\partial(f+g)(0) = \mathbb{R}$$

$$
\begin{aligned}
T : V \to W, \quad T(x) = Ax + b \\
g : E \to \mathbb{R}, \quad E \subseteq W, \quad x_0 \in T^{-1}(E) \subseteq V \\
\implies \partial g(T(x_0))(x_0) \supseteq A^* \partial g(T(x_0));
\end{aligned} \tag{8}
$$

$$
g \text{ convex}, \quad T(V) \cap \text{int}E \neq \emptyset \implies \partial g(T(x))(x_0) = A^* \partial g(T(x_0))
$$

Example:

$$
f(x) = \|Ax - b\|_1
$$

We know the subdifferential of L1 norm:

$$
\partial(\|z\|_1) = \sum_i \partial|z_1| = \sum_i \begin{cases} z_i > 0: & \{e_i\} \\ z_i < 0: & -\{e_i\} \\ z_i = 0: & e_i[-1, 1] \end{cases}
$$

Therefore using the property, we get:

$$
\partial\|Ax - b\|_1 = A^T \partial(\|\cdot\|_1)(Ax - b)
$$

Example (LASSO regression):

$$
\frac{1}{2N}\|Xw - y\|_2^2 + \lambda\|w\|_1 \to \min_w
$$

Since this is sum of 2 functions, there will be a sum of 2 subdifferentials. First term is differentiable and there is only 1 point.

$$
\partial f(w) = \frac{1}{N} X^T(Xw - y) + \lambda \sum_i \begin{cases} w_i > 0: & \{e_i\} \\ w_i < 0: & -\{e_i\} \\ w_i = 0: & e_i[-1, 1] \end{cases}
$$

Let us check whether $w = 0$ is an optimal point:

$$
0 \in \partial f(0) \implies 0 \in -\frac{1}{N} X^T y + \lambda e_i[-1, 1] \ \ \forall i \implies \left| \frac{1}{N}(X^T y)_i \right| \leq \lambda \ \ \forall i
$$

We can also write this equivalently:

$$
\boxed{0 \in \partial f(0) \iff \left\| \frac{1}{N} X^T y \right\|_\infty \leq \lambda}
$$

This is a useful criterion that helps check whether zero is already a solution to LASSO. If that is true, we need not do any optimization.

$$
\begin{aligned}
f(x) = \max(f_1(x), \ldots, f_m(x)) \implies \\
\implies \partial f(x) \supseteq \text{Conv}\left(\bigcup_{x \in I(x)} \partial f_i(x) \right) \\
I(x) := \{1 \leq i \leq m \mid f_i(x) = \max(f_1(x), \ldots, f_m(x))\};
\end{aligned} \tag{9}
$$

$$
f_1, \ldots, f_m \text{ convex}, x \in \text{int}\left(\bigcap_i E_i \right) \implies \partial f(x) = \text{Conv}\left(\bigcup_{i \in I(x)} \partial f_i(x) \right)
$$

Example (absolute value):

$$|x| = \max(x, -x)$$

$$\partial|x| = \begin{cases} x > 0: & \{1\} \\ x < 0: & \{-1\} \\ x = 0: & \text{Conv}(\{1\} \cup \{-1\}) = [-1, 1] \end{cases}$$

Example (maximum eigenvalue):

$$\lambda_{max}(A) : \mathbb{S}^n \to \mathbb{R}$$
$$\lambda_{max}(A) = \max_{x:\|x\|=1} \langle Ax, x\rangle \qquad I(A) = \{x \in \mathbb{R}^n \mid \langle Ax, x\rangle = \lambda_{max},\ \|x\| \leq 1\}$$

Since $\langle Ax, x\rangle = \text{tr}(Axx^T)$ is differentiable w.r.t. A and has gradient xx^T, we can now write:

$$\partial\lambda_{max}(A) = \text{Conv}\left(\bigcup_{x\in I(A)} xx^T\right)$$

From this expression we can see that if λ_{max} has only one eigenvector, the spectral radius is differentiable.

4 Worst Example for the Subgradient Method

Consider an optimization task:

$$f(x) = \max_{1\leq i\leq k} x_i + \frac{1}{2}\|x\|_2^2 \to \min_{x\in\mathbb{R}^n}$$

where $k < n$ is some fixed number. From the addition property and "max" property we obtain:

$$\partial f(x) = x + \text{Conv}\left(\bigcup_{i\in I(x)} \{e_i\}\right)$$

Now it's easy to see that for the optimal point x_* which satisfied $0 \in \partial f(x_*)$ all the coordinates $k+1, \ldots, n$ are zero. Now, as for the first k coordinates, if we consider that some of them are negative and some are zero, we get a contradiction: $I(x)$ will contain only indices with zero coordinates, whereas first term x will be non-zero for the other set. So, in addition, they can't be zero.

This implies that all of the coordinates from 1 to k must be negative. Furthermore, we must choose them so that $x + \text{Conv}\left(\bigcup_{1\leq i\leq k}\{e_i\}\right) = 0$. The only way how this is possible is:

$$x_{*i} = -\frac{1}{k}, \quad i = 1, \ldots, k$$
$$x_{*i} = 0, \quad i = k+1, \ldots, n$$

And the optimal value is:

$$f(x_*) = -\frac{1}{k} + \frac{1}{2}\sum_{i=1}^{k}\frac{1}{k^2} = -\frac{1}{2k}$$

Let us perform the subgradient method starting from point $x_0 = 0$. At each iteration l, we choose the subgradient in the following way:

$$z_l \in \partial f(x_l) = x_l + e_l, \quad i = \arg\min I(x_l)$$

We obtain the following trajectory:

$$x_0 = 0$$
$$x_1 = x_0 - \alpha_0 z_0 = x_0 - \alpha_0 e_1 = -\alpha_0 e_1$$
$$x_2 = x_1 - \alpha_1 z_1 = -\alpha_0 e_1 - \alpha_1 e_2$$
$$\ldots$$

For iteration $j < k$ we have:

$$f(x_j) - f_* = \max_{1 \le i \le k} x_i + \frac{1}{2}\|x\|^2 + \frac{1}{2k} = 0 + \frac{1}{2}\|x\|^2 + \frac{1}{2k} \ge \frac{1}{2k}$$

Since we have:

$$R = \|x_0 - x_*\|_2 = \sqrt{\sum_{i=1}^{k} \frac{1}{k^2}} = \frac{1}{\sqrt{k}}$$

And L is some constant greater than 1, we have the following convergence speed:

$$\frac{RL}{2\sqrt{k}} = \frac{L}{2k}$$

Since we can choose arbitrarily large dimension n for the task, this is one of the worst-case scenarios where we have the worst available convergence rate.

5 Polyak Optimal Step Size

From the convergence speed derivation we had:

$$\|x_{k+1} - x_*\|^2 \le \|x_k - x_*\|^2 - 2\alpha_k(f(x_k) - f_*) + \alpha_k^2\|z_k\|^2$$

Now, is we want to obtain the optimal step size, we need to minimize this upper bound. This gives a solution:

$$\alpha_k = \frac{f(x_k) - f_*}{\|z_k\|^2}$$

Such step size is called **Polyak optimal step size**. We obtain:

$$\|x_{k+1} - x_*\|^2 \le \|x_k - x_*\|^2 - 2\frac{(f(x_k) - f_*)^2}{\|z_k\|^2} + \frac{\|z_k\|^2(f(x_k) - f_*)^2}{\|z_k\|^4} = \|x_k - x_*\|^2 - \frac{(f(x_k) - f_*)^2}{\|z_k\|^2}$$

So, we get:

$$(f(x_k) - f_*)^2 \le \|z_k\|^2 \left(\|x_k - x_*\|^2 - \|x_{k+1} - x_*\|^2\right)$$

Using the Lipschitz property: $\|z_k\| \le L$ and summing up over first k iterations, we get:

$$(k+1)(f(\overline{x}_k) - f_*)^2 \le \sum_{i=0}^{k}(f(x_i) - f_*)^2 \le L^2\left(\|x_0 - x_*\|^2 - \|x_{k+1} - x_*\|^2\right) \le L^2R^2$$

Thus, we finally get the optimal convergence rate:

$$f(\overline{x}_k) - f_* \le \frac{LR}{\sqrt{k+1}}$$